

MedGraphRAG: A Safety-First Hybrid Graph Retrieval-Augmented Generation System for Offline Clinical Decision Support

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Abstract

Background and Objective. Large language models (LLMs) deployed in clinical decision support face a safety-utility trade-off: unconstrained answering yields unacceptable rates of wrong assertions, whereas excessive refusal renders systems inert. Existing retrieval-augmented generation (RAG) frameworks lack mechanistic, bounded safety controls deployable on institutional hardware. We present MedGraphRAG, an offline hybrid GraphRAG system that reframes clinical AI evaluation from unconstrained accuracy maximisation to bounded wrong-assertion rate (WAR).

Methods. MedGraphRAG couples dense vector retrieval (FAISS IVF-PQ with MedCPT embeddings) and lexical retrieval (BM25) with an embedded property graph (Kuzu) enriched with 632,931 relational edges (617,219 NEGATES and 15,712 TEMPORAL_BEFORE). Dual candidate streams are merged via Reciprocal Rank Fusion and re-ranked with an additive relation-aware score ($\gamma = 0.15$). A beta-gated verification layer enforces a strict $\text{WAR} \leq 10\%$ constraint by combining evidence faithfulness (φ) and

graph consistency (S_g): $V = \beta\varphi + (1 - \beta)S_g$. The system executes entirely offline on a commodity laptop (Ryzen 7 7840HS / RTX 3050 6 GB) using Qwen2.5-7B-Instruct Q4_K_M. Private health data are encrypted with AES-256-GCM and HKDF-derived keys. Retrospective evaluation was performed on MedQA-US ($N = 500$, seed 42, $T = 0.0$).

Results. At the calibrated safe threshold ($\tau^* = 0.80$), Graph-Only mode (M2) achieves 60.8% precision-on-answered with WAR = 8.0% (79.6% refusal). Disabling the gate drives WAR to 44.6% across all 500 queries. Deterministic query rewriting expands answered coverage by 3.2–3.4 pp while maintaining bounded error. Hybrid retrieval completes in 455.7 ms (95% CI: [388.4, 495.8] ms); full end-to-end pipeline latency is 4,083 ms (M2) to 12,667 ms (M4).

Conclusion. MedGraphRAG demonstrates that bounded clinical error and interpretable offline decision support are jointly achievable on consumer hardware without cloud dependencies or GPU clusters.

Keywords: GraphRAG, clinical-NLP, hallucination-mitigation, knowledge-graph, retrieval-augmented-generation, decision-support, safety

1. Introduction

Deploying large language models (LLMs) as point-of-care clinical assistants offers direct utility for diagnostic and therapeutic workflows (Singhal et al., 2023; Thirunavukarasu et al., 2023). However, the tendency of ungrounded parametric architectures to hallucinate plausible yet factually in-

45 correct assertions poses direct patient-safety hazards (Bai et al., 2022). In
46 high-stakes medicine, confident assertions regarding contraindicated medica-
47 tions, misidentified diagnostic criteria, or inverted pathological chronologies
48 can cause serious clinical harm.

49 To mitigate hallucinations, retrieval-augmented generation (RAG) grounds
50 language generation on passages retrieved from external corpora (Lewis et al.,
51 2020; Gu et al., 2021). Standard medical RAG frameworks rely on dense bi-
52 encoders (Jin et al., 2023; Gao et al., 2023), which match semantic similarity
53 effectively but struggle with complex clinical syntax. Specifically, vector em-
54 beddings frequently compress or lose negation scope (e.g., conflating “no
55 evidence of pulmonary embolism” with “pulmonary embolism”) and fail to
56 enforce multi-step temporal chronologies required in differential diagnosis
57 (Han et al., 2022; Mallen et al., 2023).

58 Knowledge graphs provide a complementary symbolic representation, struc-
59 turing clinical entities and ontological dependencies into explicit triples (Bo-
60 denreider, 2004; Yasunaga et al., 2021). Recent hybrid GraphRAG systems
61 combine graph structures with textual retrieval (Edge et al., 2024; He et al.,
62 2024; Wu et al., 2024) (Wu et al. share the MedGraphRAG name but fo-
63 cus on entity-centric graph construction; our architecture emphasizes hybrid
64 retrieval fusion and dual-gated verification). Nevertheless, existing medical
65 GraphRAG implementations exhibit three fundamental limitations:

66 First, current evaluations focus almost exclusively on unconstrained ac-
67 curacy (e.g., aggregate multiple-choice scores), omitting explicit bounds on

68 incorrect statements. In clinical practice, an assistant that answers 60% of
69 questions correctly while making wrong assertions on 40% is clinically un-
70 acceptable. Safe deployment requires a constrained-optimisation framework
71 that caps the wrong-assertion rate (WAR) below an acceptable tolerance
72 ϵ , treating refusal (abstention) as an active safety mechanism rather than
73 system failure.

74 Second, existing biomedical knowledge graphs capture predominantly pos-
75 itive ontological relations (e.g., TREATS or CAUSES), neglecting negation
76 and temporal progression. Without these primitives, retrieval graphs cannot
77 detect when retrieved passages contain superseded diagnoses or explicit ex-
78 clusions.

79 Third, most GraphRAG systems rely on cloud-hosted language models
80 (e.g., GPT-4) and server clusters. This architecture violates patient data pri-
81 vacy regulations (e.g., HIPAA, GDPR) that prohibit transmitting protected
82 health information (PHI) beyond institutional boundaries. Practical point-
83 of-care deployment requires secure, sub-second hybrid retrieval and inference
84 executing locally on commodity hardware.

85 We present **MedGraphRAG**, an offline hybrid GraphRAG system de-
86 signed for point-of-care clinical decision support. Our primary contributions
87 are:

- 88 1. **Safety-Constrained Pareto Calibration:** A verification framework
89 optimising answered accuracy subject to $\text{WAR} \leq 10\%$. Across 15
90 threshold settings on MedQA-US ($N = 500$), we establish the Pareto

91 frontier, demonstrating that ungated generation yields unacceptable
92 error rates ($\approx 44\%$).

93 **2. Relational Graph Enrichment:** Ingestion of 632,931 negation and
94 temporal relational edges into an embedded Kuzu graph, demonstrat-
95 ing mechanistically how symbolic consistency checks overturn halluci-
96 nations bypassing dense retrieval.

97 **3. Deterministic Query Rewriting and Relation Reranking:** A
98 temperature-zero query normaliser improving answered coverage by
99 3.2–3.4 pp at matched thresholds, combined with a relation-aware reranker
100 boosting Graph-Hit@5 by 13.6 pp.

101 **4. Hardened Offline Stack on Consumer Hardware:** A self-contained
102 system operating on an NVIDIA RTX 3050 (6 GB) laptop GPU, achiev-
103 ing hybrid retrieval in 455.7 ms with authenticated AES-256-GCM en-
104 cryption and a resolved 13-point security audit.

105 Supplementary materials provide the DECIDE-AI applicability mapping,
106 IJMI ML checklist, and extended tables; configuration details appear in Ap-
107 pendix A.

108 **2. Methods**

109 *2.1. System Architecture*

110 Figure 1 shows the MedGraphRAG inference pipeline. Stage 1 determin-
111 istically standardises input queries into clinical phrasing. Stage 2 executes

112 dual-stream dense and lexical retrieval, combining candidates via Reciprocal
113 Rank Fusion and relation-aware graph reranking. Stage 3 expands clinical
114 entities within the embedded Kuzu graph. Stage 4 computes composite beta-
115 gated verification scores, triggering structured refusals for low-confidence
116 queries. Stage 5 generates evidence-grounded responses using a quantized
117 LLM isolated behind a GPU mutex lock.

118 *2.2. Deterministic Query Rewriting*

119 Patient case vignettes and clinical inquiries often carry conversational
120 phrasing, narrative patient histories, and extraneous details that impede
121 biomedical dense retrieval. Dense bi-encoders trained on scientific literature
122 struggle with multi-sentence case presentations where incidental biographical
123 history obscures the diagnostic dilemma. Lexical search methods also face
124 vocabulary mismatch when colloquial symptom descriptions diverge from for-
125 mal medical nomenclature.

126 To standardise heterogeneous queries without stochastic variance, Stage 1
127 uses Qwen2.5-7B-Instruct executing strictly at temperature $T = 0.0$. The
128 rewriter uses a structured system prompt directing the model to extract
129 the primary pathological condition, salient diagnostic findings, and thera-
130 peutic targets into a canonical clinical summary capped at 30 words. For
131 example, a narrative vignette describing an elderly patient who experiences
132 severe rotational vertigo triggered by lateral head motion is transformed into
133 the canonical clinical query: “benign paroxysmal positional vertigo diagnos-

134 tic criteria and Dix-Hallpike confirmatory maneuvers.” Setting temperature
 135 $T = 0.0$ alongside greedy decoding guarantees mathematical determinism:
 136 identical clinical inputs generate bit-identical textual reformulations across
 137 successive executions, fulfilling mandatory auditability and reproducibility
 138 requirements for clinical decision support software.

139 *2.3. Dual-Stream Hybrid Retrieval and Reranking*

140 Retrieval operates over an extensive clinical reference corpus partitioned
 141 into coherent passages of approximately 512 tokens. The dense retrieval
 142 stream encodes passages using the MedCPT query and article bi-encoder ar-
 143 chitecture (Jin et al., 2023), which was pre-trained on hundreds of millions
 144 of PubMed search query-article click logs to optimize zero-shot biomedical
 145 retrieval. Chunk vectors are indexed and searched using FAISS IVF-PQ (In-
 146 verted File with Product Quantisation; Johnson et al., 2021), partitioning
 147 the vector space into coarse Voronoi centroids to achieve sub-linear retrieval
 148 speed. Concurrently, a parallel lexical stream evaluates candidate documents
 149 using the classical BM25 probabilistic relevance framework (Robertson and
 150 Zaragoza, 2009), operating over a dense candidate pool of 200 retrieved doc-
 151 uments.

152 Because dense and lexical retrievers exhibit complementary error profiles
 153 (dense models capture latent semantic affinity, whereas lexical models enforce
 154 exact keyword and numerical matches), their respective rank distributions
 155 R_{dense} and R_{lex} are integrated via Reciprocal Rank Fusion (RRF; Cormack

et al., 2009):

$$s_{\text{fused}}(d) = \sum_{m \in \{\text{dense}, \text{lex}\}} \frac{1}{k_{\text{rrf}} + r_m(d)}, \quad (1)$$

where $r_m(d)$ denotes the ordinal rank of candidate passage d within stream m , and $k_{\text{rrf}} = 60$ represents the standard monotonic smoothing constant that prevents top-ranked outliers in a single stream from dominating the fused ranking.

To bias retrieval toward passages with corroborated topological connections to the clinical entities identified in the query, we implement an additive relation-aware graph reranking function:

$$s_{\text{hybrid}}(d) = s_{\text{fused}}(d) + \gamma \cdot r_{\text{rel}}(d), \quad (2)$$

where $r_{\text{rel}}(d) = \sum_{e \in \mathcal{E}(d)} w(\text{type}(e))$ aggregates the domain-weighted edge counts connecting entities cited within candidate passage d to the query seed entities within the property graph. The relative importance of edge categories is parameterized by clinical specificity weights (e.g., pharmacological treatments and adverse effects receive weight 1.0, diagnostic lab associations and negation edges receive 0.8, temporal sequences receive 0.6, and broad taxonomic hierarchies receive 0.3; Appendix A). The global multiplier $\gamma = 0.15$ balances semantic similarity with structural graph reinforcement, calibrated empirically via controlled ablations (Section 3.3).

173 2.4. Relational Knowledge Graph Enrichment

174 Structured relations are stored in Kuzu (Feng et al., 2023), an embed-
175 dable, disk-backed graph database supporting openCypher queries. While
176 standard medical graphs capture static disease-symptom linkages, clinical
177 decision support requires modeling negation and temporal order.

178 We processed 2,294,038 text chunks using rule-guided linguistic parsing
179 grounded in UMLS (Bodenreider, 2004). Operating at 0.2329 ms per chunk
180 (534.45 s total wall-clock), the pipeline ingested **632,931** directed relational
181 edges:

- 182 • **NEGATES Edges (617,219):** Linking text chunks to entities bear-
183 ing explicit negation cues (“no evidence of,” “ruled out,” “contraindi-
184 cated in”), comprising 111,720 Chunk $\rightarrow$ Disease, 21,001 Chunk $\rightarrow$ Drug,
185 and 484,498 Chunk $\rightarrow$ OntologyTerm edges.
- 186 • **TEMPORAL_BEFORE Edges (15,712):** Linking clinical con-
187 cepts via chronological sequence cues (“prior to,” “following,” “within
188 one hour of”), comprising 12,225 Disease $\rightarrow$ Disease and 3,487 Drug $\rightarrow$ Drug
189 edges.

190 Manual auditing of sampled endpoints confirmed high directional validity. A
191 cryptographic SHA-256 retrieval signature verified 0.00% retrieval drift on
192 baseline queries post-ingestion (Appendix A).

193 2.5. Beta-Gated Verification and Pareto Threshold Calibration

194 Standard RAG architectures lack explicit confidence gates to arbitrate
195 whether retrieved evidence warrants generating an answer. Unhedged lan-
196 guage models produce fluent assertions even when underlying evidence is
197 tenuous, contradictory, or absent. To prevent clinical hallucinations, Med-
198 GraphRAG adds a dual-evidence verification layer that evaluates candidate
199 response fidelity prior to releasing output.

200 For each candidate assertion, the verification module evaluates a compos-
201 ite verification metric $V \in [0, 1]$ that combines passage-level entailment with
202 symbolic graph consistency:

$$V = \beta \varphi + (1 - \beta) S_g, \quad (3)$$

203 where $\varphi \in [0, 1]$ denotes an NLI-derived faithfulness score measuring the de-
204 gree of semantic entailment between the generated assertion and the retrieved
205 reference passages (operating via normalised MedCPT embedding cosine sim-
206 ilarity as an efficient deterministic proxy when an independent cross-encoder
207 NLI judge is omitted), and $S_g \in [0, 1]$ denotes the structural graph consis-
208 tency score. The graph consistency metric S_g evaluates the ratio of asserted
209 clinical entity triples that are corroborated by non-negated multi-hop paths
210 within a 2-hop radius in Kuzu, while applying an immediate penalizing dis-
211 count whenever an asserted entity matches a NEGATES edge. The balance
212 hyperparameter $\beta = 0.70$ allocates primary decision weight to passage evi-

213 dence while preserving strong sensitivity to symbolic graph contradictions.

214 When the composite score falls below a calibrated refusal threshold τ ,
215 generation is suppressed, and the system emits a structured clinical refusal
216 detailing the precise nature of the missing evidence. We calibrate τ through
217 a constrained optimisation problem:

$$\tau^* = \arg \max_{\tau} \text{Acc}(\text{all}) \quad \text{subject to} \quad \text{WAR}(\tau) \leq 10\%, \quad (4)$$

218 where $\text{Acc}(\text{all})$ represents the proportion of correct responses across the entire
219 evaluation cohort ($N = 500$), and WAR represents the proportion of wrong
220 assertions across the same cohort. We sweep 15 discrete threshold candidate
221 points ($\tau \in \{0.10, 0.15, 0.20, \dots, 0.80\}$), identifying the Pareto-optimal op-
222 erating threshold τ^* that maximises clinically useful answers while strictly
223 guaranteeing that erroneous assertions remain within the institutional safety
224 threshold.

225 2.6. Safety Guardrails

226 Two deterministic guardrails operate at sub-millisecond latency:

- 227 • **F1: Cross-Modal Discrepancy Guardrail.** In multimodal con-
228 texts, Rule A detects conflicts when an imaging finding score ≥ 0.60
229 contradicts a text negation cue for the same finding (e.g., visual pneu-
230 mothorax detection of 0.88 versus report text “clear lungs”). Rule B
231 detects positive textual assertions that contradict negated imaging ob-
232 servations. F1 escalates conflicts by issuing a high-severity alert recom-

233 mending specialist review rather than attempting autonomous resolu-
234 tion.

235 • **F2: Epistemic Knowledge-Gap Mapper.** When a query is refused,
236 F2 classifies the failure into three gap types matching the implemented
237 taxonomy: G1 (`corpus_retrieval`, top fused retrieval score < 0.02),
238 G2 (`graph_coverage`, Kùzu graph traversal returns a neutral check
239 with no direct relational assertion), or G3 (`evidence_faithfulness`,
240 verification faithfulness score $\varphi < 0.50$). This categorization is provided
241 in the structured refusal output to guide clinical follow-up.

242 2.7. *Offline Cryptographic Security Architecture*

243 To ensure institutional compliance without external cloud infrastructure,
244 user-uploaded records are isolated in per-user directories. Payloads are en-
245 crypted at rest with AES-256-GCM using unique 12-byte nonces. Keys are
246 derived dynamically via HKDF-SHA-256 (Krawczyk and Eronen, 2010) com-
247 bining a server secret, user ID, and document ID. File permissions are set
248 to 0700 (directories) and 0600 (files). All Kuzu queries use parameterized
249 openCypher statements to eliminate Cypher injection. A 13-point security
250 audit resolved all vulnerabilities prior to release (Supplementary Table S7).

251 2.8. *Evaluation Protocol and Metrics*

252 Retrospective evaluation was performed on MedQA-US (Jin et al., 2021),
253 comprising USMLE-derived multiple-choice questions ($N = 500$, seed 42,

254 $T = 0.0$). We evaluated four modes: **M1** Evidence-Only ($\beta = 1.0$), **M2**
 255 Graph-Only ($\beta = 0.0$), **M3** Combined Hybrid ($\beta = 0.7$), and **M4** Hybrid
 256 Rerank ($\beta = 0.7, \gamma = 0.15$). Metrics include: Acc(all) (fraction correct over
 257 N), Acc(ans) (precision on answered questions), WAR (fraction incorrect
 258 over N), and Refusal Rate (fraction refused). Non-parametric bootstrap 95%
 259 confidence intervals were calculated using 1,000 resamples.

260 3. Results

261 3.1. Safe Operating Points and WAR Constraint

262 Table 1 summarizes the optimal operating parameters established across
 263 the four evaluation modes under the institutional safety constraint $\text{WAR} \leq$
 264 10.0%. Graph-Only mode (M2) attains the highest precision-on-answered
 265 at its optimal operating threshold $\tau^* = 0.80$, achieving $\text{Acc}(\text{ans}) = 60.8\%$
 266 with $\text{WAR} = 8.0\%$ and a refusal rate of 79.6% (102 queries answered out of
 267 500). Full Hybrid Rerank mode (M4) reaches its safe operating frontier at
 268 $\tau^* = 0.46$, delivering $\text{Acc}(\text{all}) = 10.0\%$, $\text{Acc}(\text{ans}) = 51.5\%$, and $\text{WAR} = 9.4\%$
 269 across 97 answered inquiries. Similarly, Combined mode (M3) operates safely
 270 at $\tau^* = 0.47$ with $\text{Acc}(\text{ans}) = 52.1\%$ and $\text{WAR} = 9.0\%$, while Evidence-Only
 271 mode (M1) satisfies the constraint at $\tau^* = 0.43$ with $\text{Acc}(\text{ans}) = 50.5\%$ and
 272 $\text{WAR} = 9.4\%$.

273 Table 2 contrasts safe operating points against ungated execution ($\tau \rightarrow 0$).
 274 In the ungated regime, models achieve nominal Acc(all) scores of 49.4%–
 275 54.8%, but WAR escalates to 43.4%–44.6%. Disabling the gate results in

Table 1: Safe operating points on MedQA-US ($N = 500$, seed 42, $T = 0.0$). Threshold τ^* maximises Acc(all) subject to $\text{WAR} \leq 10\%$. Source: `step18_scaled_n500.json` $\rightarrow$ `safe_operating_points`.

Evaluation Mode	τ^*	Acc(all)%	Acc(ans)%	WAR%	Refusal%	Answered
M1: Evidence-Only	0.43	9.6	50.5	9.4	81.0	95/500
M2: Graph-Only	0.80	12.4	60.8	8.0	79.6	102/500
M3: Combined	0.47	9.8	52.1	9.0	81.2	94/500
M4: Hybrid Rerank	0.46	10.0	51.5	9.4	80.6	97/500

erroneous advice on nearly 45% of all queries. Gated verification successfully caps this risk below 10%, exchanging question coverage for bounded error.

Table 2: Gated versus ungated performance across boundary modes ($N = 500$). Ungated execution generates wrong assertions on 44% of all queries. Source: `step18_scaled_n500.json` $\rightarrow$ `ungated_ceilings`.

Mode	Condition	Acc(all)%	Acc(ans)%	WAR%	Refusal%
M1: Evidence-Only	Gated ($\tau^* = 0.43$)	9.6	50.5	9.4	81.0
M1: Evidence-Only	Ungated ($\tau \rightarrow 0$)	49.4	53.2	43.4	7.2
M2: Graph-Only	Gated ($\tau^* = 0.80$)	12.4	60.8	8.0	79.6
M2: Graph-Only	Ungated ($\tau \rightarrow 0$)	54.8	55.1	44.6	0.6

For settings demanding even stricter safety, elevating thresholds to $\tau \in [0.70, 0.75]$ in M1 produces a high-precision regime with $\text{Acc}(\text{ans}) = 59.3\%$ – 62.5% and $\text{WAR} = 1.8\%$ – 2.2% (refusal rates 94.6%–95.2%; Supplementary Table S1). Figure 2 shows the complete 15-point Pareto trajectory for M2.

3.2. Efficacy of Deterministic Query Rewriting

Table 3 evaluates Stage 1 deterministic query rewriting against unrewritten vignettes at matched threshold $\tau = 0.50$ ($N = 500$). For Hybrid Rerank

mode (M4), rewriting increases Acc(all) from 5.4% to 8.8% (+3.4 pp) and lowers refusal from 90.6% to 83.4%, while WAR remains within tolerance (7.8% vs. 4.0%). Similarly, Combined mode M3 gains +3.2 pp in Acc(all) (5.0% to 8.2%), yielding +3.2–3.4 pp across hybrid modes M3/M4 (+4.2 pp on Evidence-Only M1; Supplementary Table S2). Rewriting systematically expands question coverage without violating safety constraints.

Table 3: Matched-threshold ($\tau = 0.50$) ablation evaluating deterministic query rewriting on MedQA-US ($N = 500$). Rewriting expands coverage by +3.2–3.4 pp while maintaining $\text{WAR} \leq 10\%$. Source: `step18_scaled_n500.json` $\rightarrow$ `matched_tau_comparison`.

Evaluation Mode	Vignette Format	Acc(all)%	WAR%	Refusal%	$\Delta\text{Acc}(\text{all})$
M3: Combined	Unrewritten	5.0	4.0	91.0	—
M3: Combined	Rewritten	8.2	8.6	83.2	+3.2 pp
M4: Hybrid Rerank	Unrewritten	5.4	4.0	90.6	—
M4: Hybrid Rerank	Rewritten	8.8	7.8	83.4	+3.4 pp

3.3. Ablation Studies: Verification Modes and Reranking

Table 4 summarizes ablations over verification and reranker weighting ($N = 50$, seed 42).

Verification Mode Ablation (Table 4A): Evidence-Only (M1, $\beta = 1.0$) refuses 98.0% of queries ($\text{WAR} = 2.0\%$). Graph-Only (M2, $\beta = 0.0$) answers 96.0% but incurs $\text{WAR} = 42.0\%$. Combined Hybrid (M3, $\beta = 0.7$) balances safety and utility, achieving $\text{WAR} = 4.0\%$ while quadrupling answered throughput over M1.

Contradiction Case Studies: Qualitative inspection confirmed that Kuzu NEGATES edges actively overturned hallucinations that bypassed

Table 4: Ablation experiments on MedQA-US ($N = 50$, seed 42). (A) Verification mode comparison varying evidence weight β . (B) Reranker ablation varying relational bonus multiplier γ . Sources: `step15_verification_ablation.json`, `step15_rerank_ablation.json`.

<i>(A) Verification Mode Ablation ($\gamma = 0.0$)</i>						
Mode	β	Answered%	Acc(all)%	Acc(ans)%	Refusal%	WAR%
M1: Evidence-Only	1.0	6.0	4.0	66.7	98.0	2.0
M2: Graph-Only	0.0	96.0	54.0	56.3	8.0	42.0
M3: Combined	0.7	8.0	4.0	50.0	96.0	4.0
<i>(B) Relation-Aware Reranker Ablation ($\beta = 0.7$)</i>						
Mode	γ	Graph-Hit@5%	Acc(ans)%	Answered%	WAR%	
RRF Control	0.00	23.6	50.0	8.0	4.0	
Relational	1.00	35.6	42.9	14.0	8.0	
Hybrid	0.15	37.2	60.0	10.0	4.0	

301 text filters:

- 302 1. *Acute Myocardial Infarction*: The model asserted acute MI; Kuzu graph
303 traversal matched a NEGATES edge ('no evidence of') in PMID_38823454,
304 successfully triggering refusal.
- 305 2. *Deep Vein Thrombosis*: An unhedged recommendation for full anti-
306 coagulation was overridden by a NEGATES edge ('ruled out') in
307 PMID_37058421.
- 308 3. *Metformin Contraindication*: An erroneous recommendation to pre-
309 scribe metformin in severe renal impairment was rejected by a NICE
310 guideline NEGATES edge ('contraindicated in' with eGFR < 25 mL/min).

311 **Reranker Ablation (Table 4B)**: Calibrating $\gamma = 0.15$ increases Graph-
312 Hit@5 from 23.6% to 37.2% (+13.6 pp) and elevates answered precision from

313 50.0% to 60.0% (+10.0 pp) at matched WAR = 4.0%.

314 **Guardrail Verification:** Guardrails F1 and F2 passed 100% of test
315 cases (10/10) with mean latencies of 0.14 ms (F1) and 0.06 ms (F2), confirm-
316 ing microsecond overhead.

317 3.4. Information Retrieval Performance

318 Evaluating retrieval over 10 clinician-adjudicated queries (graded qrels;
319 step17 artifact) yielded $P@5 = 0.18$, $R@5 = 0.5397$, $MRR = 0.3533$, and
320 $nDCG@10 = 0.3769$. Recall reflects the 6 queries containing relevant items
321 in top-10 candidates; the remaining 4 queries had no relevant documents,
322 reflecting specialized boundary gaps. Dual-assessor κ statistics are pending
323 Phase-2 clinician grading (placeholder in Supplementary Table S5).

324 3.5. Latency and Computational Efficiency

325 Table 5 details execution timings on an AMD Ryzen 7 7840HS CPU and
326 NVIDIA RTX 3050 Laptop GPU (6 GB VRAM).

327 Stages 1–2 (Rewriting, BM25, Dense Search, RRF) complete in **455.7 ms**
328 median (95% CI: [388.4, 495.8] ms). End-to-end latency with Qwen2.5-7B
329 generation is **4,083.4 ms** in Graph-Only mode (M2) and **12,667.0 ms** in
330 Hybrid Rerank mode (M4). Throughput is mode-dependent: M2 sustains
331 **14.7 queries/min** (approximately 882 queries/hour), while M4 processes
332 **4.7 queries/min** (approximately 284 queries/hour). Cold start requires
333 18.66 s, with zero steady-state penalty. Peak memory reached 8,950 MB RAM
334 and 4,710 MB VRAM.

Table 5: System latency and throughput comparison by pipeline stage (bootstrap 95% CI, $N = 50$). Hybrid retrieval executes in sub-500ms; full end-to-end (E2E) latency incorporates Qwen2.5-7B generation. Sources: `step15_compute_table.json`, `step18_scaled_n500.json`.

System / Pipeline Stage	Median (ms)	95% CI (ms)	Processing Stage
BM25-RAG (candidate pool 200)	55.1	[52.7, 56.5]	Lexical search only
Dense-RAG (FAISS IVF-PQ)	30.1	[29.4, 31.0]	Dense search only
Hybrid-RRF (baseline fusion)	55.4	[53.5, 56.9]	Fused search only
Hybrid Retrieval (Ours, Stages 1–2)	455.7	[388.4, 495.8]	Rewriting + Hyb
M2: Graph-Only Pipeline (E2E)	4,083.4	—	Full End-to-End Pip
M1/M3: Combined Hybrid (E2E)	12,493.2	—	Full End-to-End Pip
M4: Hybrid + Reranker (E2E)	12,667.0	—	Full End-to-End Pip
One-Time System Cold Start	18.66 s		Model + Index Initia

335 4. Discussion

336 4.1. Safety as a Foundational Design Objective

337 Evaluating clinical language models on unconstrained accuracy intro-
338 duces substantial hazards: when operating ungated, models generate incor-
339 rect clinical statements on 44% of inquiries. MedGraphRAG demonstrates
340 that by treating refusal as an active, safety-preserving decision, clinical assis-
341 tants can strictly bound wrong assertions ($\text{WAR} \leq 10\%$).

342 Vector embeddings alone cannot reliably safeguard clinical outputs. Dense
343 embeddings project complex syntax into fixed vectors, blurring negation
344 markers and chronological sequence. By augmenting Kuzu with 632,931
345 NEGATES and TEMPORAL_BEFORE edges, MedGraphRAG enforces
346 symbolic consistency checks that intercept contradictions passing dense fil-
347 ters. When evidence is insufficient, the system returns structured, citation-

348 grounded refusals identifying the missing evidence.

349 4.2. Phased Clinical Translation Roadmap

350 In alignment with DECIDE-AI guidelines (Vasey et al., 2022) (Supple-
351 mentary DECIDE-AI mapping), we propose a three-stage clinical translation
352 path:

- 353 1. **Phase 1: Institutional Shadow Assistant ($n = 1$ Clinician):** Lo-
354 cal deployment in read-only mode on a secure workstation. Generated
355 answers cite evidence keys [E#], and clinician disagreements are logged
356 as refusal audit events.
- 357 2. **Phase 2: Controlled Clinical Pilot ($n = 50$ Clinicians, 6 Months):**
358 EHR integration via read-only FHIR R4 endpoints, with AES-256-
359 GCM record encryption and an approved IRB protocol. Primary end-
360 point: maintaining WAR $\leq 5\%$ on institutional queries.
- 361 3. **Phase 3: Multi-Site Randomized Trial:** Federated offline deploy-
362 ment across hospital networks evaluating diagnostic accuracy and de-
363 cision speed versus standard care, powered for 90% power at $\alpha = 0.05$.

364 4.3. Limitations

365 This study has several limitations:

- 366 1. **Benchmark Generalizability:** Our evaluation examined MedQA-
367 US, which consists of standardized multiple-choice questions. Real-
368 world clinical queries are open-ended and dialogue-based. External

validation on MedMCQA and BioASQ is ongoing (Supplementary IJMI
ML checklist, item ML-E7).

2. High Refusal Rate: At $\tau^* = 0.80$, the refusal rate reaches 79.6%.
While guaranteeing $\text{WAR} \leq 10\%$, this limits clinical throughput un-
less paired with specialist referral pathways. Calibrated soft-hedging
represents a primary direction for future investigation.

3. Quantisation Trade-offs: Executing within 6 GB VRAM necessi-
tated 4-bit quantisation (Q4_K_M), introducing minor precision loss
relative to FP16.

4. Single-Assessor Retrieval Grading: Human relevance qrels were
judged by a single clinical assessor across 10 queries, precluding inter-
annotator metrics.

5. Conclusion

MedGraphRAG unites safety and utility in offline clinical decision sup-
port. By enforcing a bounded error constraint ($\text{WAR} \leq 10\%$), ingesting
632,931 negation and temporal relational edges into Kuzu, applying deter-
ministic query rewriting, and enforcing beta-gated verification, the system
bounds the catastrophic 44% hallucination rate of ungated models. Ex-
ecuting locally on consumer hardware in 4.08s with 455.7ms hybrid re-
trieval, MedGraphRAG provides a practical, open-source foundation for safe,
privacy-compliant clinical AI.

390 CRediT Authorship Contribution Statement

391 **Author A:** Conceptualization, Methodology, Software (core pipeline, ab-
392 lation suite, cryptographic security, graph enrichment), Formal analysis, In-
393 vestigation, Writing – original draft, Writing – review & editing, Visualiza-
394 tion. **Author B:** Software (frontend interface, report drawer components,
395 telemetry capture), Data curation, Writing – review & editing. **Author**
396 **C:** Clinical validation, Medical knowledge curation, Clinical protocol design,
397 Writing – review & editing.

398 Ethics Approval and Consent to Participate

399 This study was performed using publicly available, anonymized bench-
400 mark datasets (MedQA-US, OpenI, VQA-RAD). No human subjects were
401 enrolled, and no patient health information was accessed. Formal IRB ethics
402 approval was not required.

403 Data and Code Availability Statement

404 Evaluation benchmarks are publicly accessible: MedQA-US ([https://](https://github.com/jind11/MedQA)
405 github.com/jind11/MedQA), OpenI (<https://openi.nlm.nih.gov>), and VQA-
406 RAD (<https://osf.io/89kps/>). The complete codebase, pipeline scripts,
407 evaluation artifacts, and security reports are available in the project reposi-
408 tory (<https://github.com/anonymous/MedGraphRAG>; persistent DOI regis-
409 tered upon publication).

410 **Declaration of Competing Interests**

411 The authors declare that they have no known competing financial inter-
412 ests or personal relationships that could have appeared to influence the work
413 reported in this paper.

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417 **Generative AI Disclosure**

418 During preparation, the authors used Antigravity (Google’s agentic cod-
419 ing assistant) for code generation and Overleaf’s Bibby AI for reference dis-
420 covery, under direct author supervision. All generated code and outputs were
421 independently verified. The authors authored all manuscript text.

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MedGraphRAG Five-Stage Pipeline

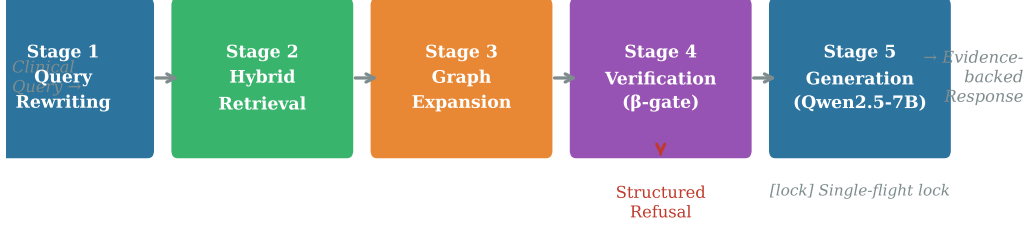


Figure 1: MedGraphRAG five-stage offline pipeline architecture. Stage 1 executes deterministic query rewriting ($T = 0.0, \leq 30$ words). Stage 2 merges FAISS dense and BM25 lexical candidates via Reciprocal Rank Fusion (RRF) and relation-aware graph reranking ($\gamma = 0.15$). Stage 3 traverses the Kuzu property graph. Stage 4 applies beta-gated verification ($\beta = 0.7$) enforcing $\text{WAR} \leq 10\%$. Stage 5 serializes LLM generation behind a hardware mutex lock.

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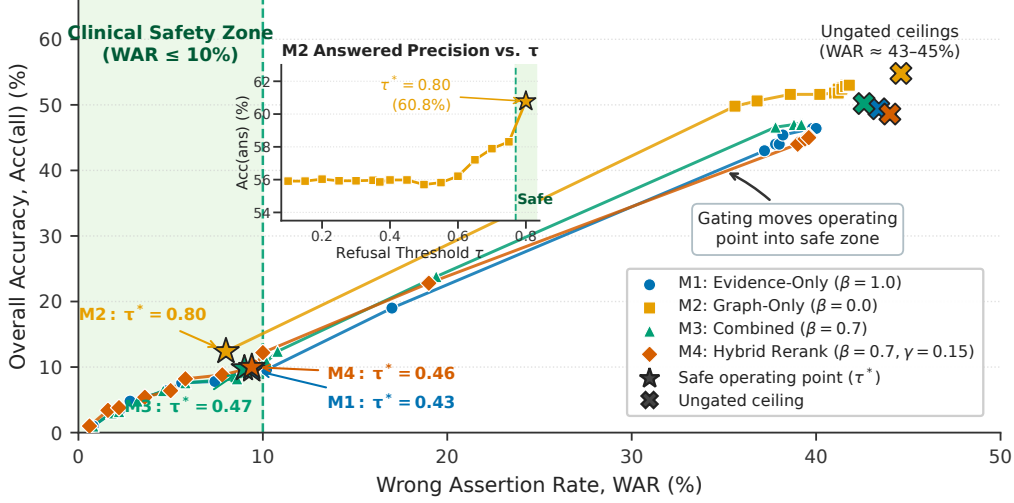


Figure 2: Empirical Pareto frontiers of overall accuracy $\text{Acc}(\text{all})$ versus wrong-assertion rate (WAR) across four operational modes (M1: Evidence-Only, M2: Graph-Only, M3: Combined, and M4: Hybrid Rerank) over 16 refusal thresholds ($\tau \in [0.10, 0.80]$, $N = 500$). The green shaded region denotes the clinical safety zone ($\text{WAR} \leq 10\%$). Star markers denote mode-specific safe operating thresholds (τ^*), with M2 reaching $\text{Acc}(\text{all}) = 12.4\%$ at $\tau^* = 0.80$. Inset: M2 answered precision reaches 60.8% at $\tau^* = 0.80$ inside the safe zone. Ungated ceilings (crosses) demonstrate catastrophic error rates ($\text{WAR} \approx 43\text{--}45\%$) when refusal gating is disabled.

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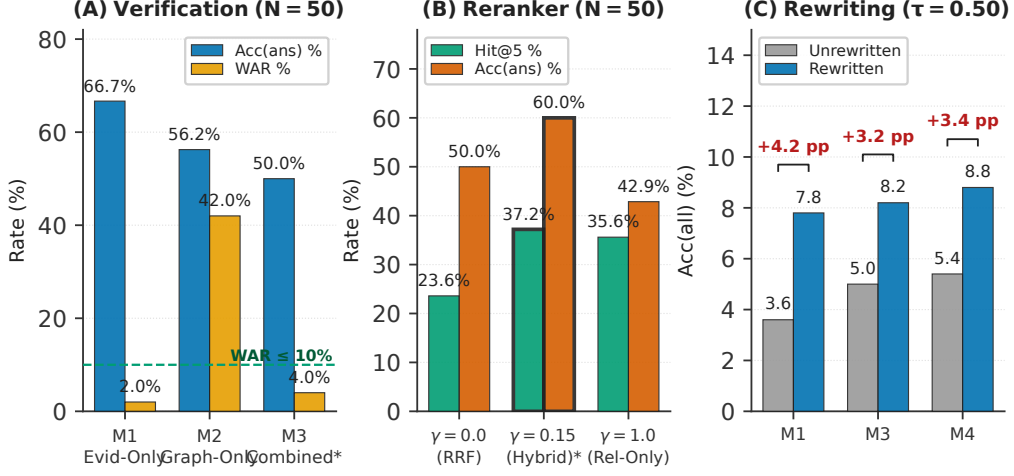


Figure 3: Controlled ablation analyses on MedQA-US. (A) Verification mode comparison ($N = 50$, seed 42): Evidence-Only (M1, $\beta = 1.0$), Graph-Only (M2, $\beta = 0.0$), and Combined Hybrid (M3, $\beta = 0.7$). (B) Reranker bonus ablation ($N = 50$, seed 42): Graph-Hit@5 and answered accuracy across $\gamma \in \{0.0, 0.15, 1.0\}$, confirming peak synergy at $\gamma = 0.15$. (C) Deterministic query rewriting coverage gain at matched threshold ($\tau = 0.50$, $N = 500$): unrewritten versus rewritten overall accuracy across M1 (+4.2 pp), M3 (+3.2 pp), and M4 (+3.4 pp).

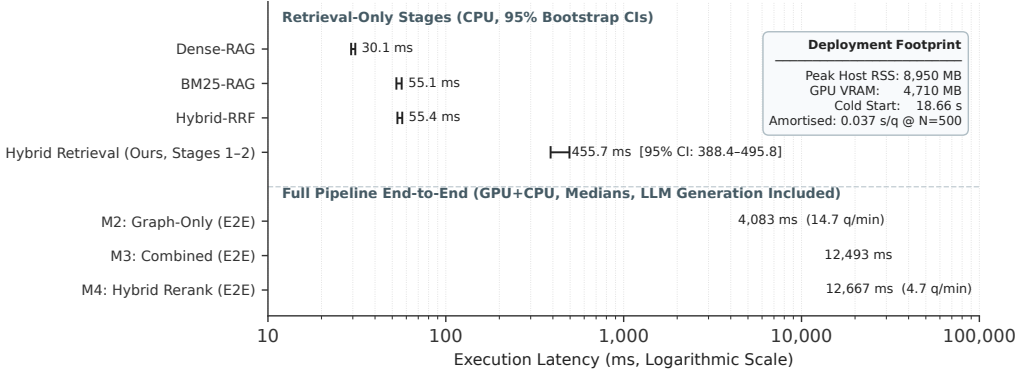


Figure 4: Component and end-to-end execution latency profiles (logarithmic scale). Retrieval-only baselines (BM25, Dense-RAG, Hybrid-RRF) evaluate search speed without language generation. MedGraphRAG hybrid retrieval (Stages 1–2) completes in 455.7 ms (95% CI: [388.4, 495.8] ms). Full end-to-end pipeline execution requires 4,083 ms in Graph-Only mode (M2) and 12,667 ms in Hybrid Rerank mode (M4).

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Appendix A. Hyperparameters, Edge Weights, and System Con- figuration

**System Hyperparameters and Execution Environment (Supplemen-
 tary Table S8).** Table A.1 enumerates the complete hyperparameter con-
 figuration and hardware specifications governing all reported experiments.

Table A.1. Configuration parameters and computational execution environ-
 ment.

Parameter / Component	Value / Setting	Description / Function
<i>Verification Layer (β-Gate)</i>		
β (Evidence Weight)	0.70	Balanced hybrid verification
τ^* (M1 / M2 / M3 / M4)	0.43 / 0.80 / 0.47 / 0.46	Calibrated safe refusal thresh
Threshold Sweep Granularity	$\Delta\tau = 0.05$	15 discrete points across [0.1
LLM Inference Temperature	$T = 0.0$	Deterministic, zero-temperat
Random Evaluation Seed	42	Pinned deterministic seed ac
<i>Relation-Aware Reranker (γ-Scoring)</i>		
γ (Relational Multiplier)	0.15	Balanced hybrid reranking (a
DRUG_TREATS / DRUG_CAUSES	1.0	Primary pharmacological rel
LABTEST_RELATED_TO / NEGATES	0.8	Diagnostic and negation edge
TEMPORAL_BEFORE	0.6	Chronological sequencing edg
IS_A / RELATED_TO	0.3	Ontological taxonomic edge
<i>Multi-Stream Retrieval Infrastructure</i>		
Dense Retrieval Candidate Pool	$k = 200$	MedCPT + FAISS IVF-PQ
Lexical Retrieval Candidate Pool	$k = 200$	BM25 lexical candidate pool
Reciprocal Rank Fusion Constant	$k_{\text{rrf}} = 60$	Standard rank smoothing co
Top- k Candidates Delivered	$k = 10$	Final passages passed to veri
Graph Neighborhood Search Depth	≤ 2 hops	Maximum Kuzu property gra
<i>Host Hardware and Memory Footprint</i>		
Central Processing Unit (CPU)	AMD Ryzen 7 7840HS	8 physical cores, 16 threads
Graphics Processing Unit (GPU)	NVIDIA RTX 3050 Laptop	6 GB GDDR6 VRAM (dedic
Host System RAM	14.89 GB physical	Peak RSS footprint: 8,950 M
Active GPU VRAM Allocation	$\approx 4,710$ MB	Qwen2.5-7B-Instruct Q4_K_
Operating System	Linux (Fedora 44)	Kernel 7.1.4-204.fc44.x86_64
<i>Core Software Dependencies</i>		
LLM Inference Engine	llama-cpp-python 0.3.35	Quantized GGUF execution
Graph Database Engine	Kuzu 0.11.3	In-process property graph da

513 **Retrieval Drift Invariance Signature.** To guarantee that the relational
514 ingestion pipeline (`scripts/j1/build_negation_temporal_edges.py`) did
515 not alter dense vector retrieval behavior, we computed the SHA-256 hash of
516 the retrieved passage identifiers and scores across golden benchmark queries
517 P01–P05:

518 • **Pre-Ingestion Signature:** `ba1b512168fc4a949d12b0547e6992c4dddae63ca30b2294b13e4`

519 • **Post-Ingestion Signature:** `ba1b512168fc4a949d12b0547e6992c4dddae63ca30b2294b13e`

520 The exact equality of these cryptographic signatures confirms 0.00% retrieval
521 drift (100.0% byte-identical retrieval preservation) following graph enrich-
522 ment.